

Homework 1

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Problem 1

Let $\{x_n\}$ be a sequence in $\mathbb{R}^m$ with $\|x_{n+1} - x_n\| \leq \frac{1}{n^2}$ for all $n \geq 1$
Then for $k > n$

$$\begin{aligned}\|x_k - x_n\| &\leq \|x_k - x_{k-1}\| + \dots + \|x_{n+1} - x_n\| \\ &\leq \frac{1}{(k-1)^2} + \dots + \frac{1}{n^2} \\ &\leq \frac{k-n}{n^2} \\ &\leq \frac{1}{k-1}\end{aligned}$$

As $k \rightarrow \infty$, we now have $\|x\|$